

Establish AI Provider Connection for the Chatbot

Task Objective

Setup and verification of the chatbot integration for the designated AI vendor. It involves checking the current AI vendor configuration, setting up environment variables, making sure that requests can be made via the current /api/ai/chat endpoint, and troubleshooting any problems related to the AI vendor connection.

AI Provider Configuration

The YAF backend provides support for various AI providers via the current provider routing framework:

- Mock
- OpenAI
- xAI
- Pinecone Assistant

As per the testing in Sprint 2, the Pinecone Assistant is chosen as the configured AI provider.

The above-mentioned provider selection was done via the following:

AI_PROVIDERS=pinecone

The required configuration details for Pinecone were configured in the local .env file as follows:

PINECONE_API_KEY=<configured>

PINECONE_ASSISTANT_NAME=travelassistant

PINECONE_ASSISTANT_HOST=<configured>

PINECONE_MODEL=gpt-4o

PINECONE_API_VERSION=2025-10

Existing Chatbot Flow Verification

The current bot code was analysed to ensure that there were no modifications needed for the API endpoint.

The chatbot request flow is using:

- /api/ai/chat endpoint
- AI Handler
- AI Orchestrator
- Failover Router with staggered fails
- AI provider configured

The current flow was used for Pinecone testing.

Local Environment Verification

The backend was successfully started locally using:

```
go run ./cmd/api
```

The startup logs confirmed that Pinecone was selected:

ai provider: pinecone

Server running on :8080

The application health endpoint was also tested:

curl <http://localhost:8080/health>

The endpoint returned:

HTTP 200 OK

confirming that the backend was running correctly.

Chatbot Request Verification

An endpoint was used for sending a request from the chatbot:

```
Invoke-RestMethod -Method POST `
```

```
-Uri "http://localhost:8080/api/ai/chat" `
```

```
-ContentType "application/json" `
```

```
-Body '{"content":"What accessibility services are available at Sydney Airport?"}'
```

It was successfully delivered to the Pinecone provider.

The backend logs showed:

```
ai: provider attempt started provider=pinecone
```

Nevertheless, the following reply was received from Pinecone:

```
invalid chat request:
```

No files found. If files were uploaded, they may still be processing

Thus, it was clear that:

- The Pinecone API key was successfully recognized
- The Pinecone Assistant connection was established
- The chatbot request was successfully transferred to Pinecone
- The only problem left was the lack of uploaded knowledge files in the Assistant

Issue Resolution

This problem was because the Pinecone Assistant did not have any sources of knowledge.

A YAF Accessibility Knowledge document was added to the Pinecone Assistant as a source of knowledge.

Example knowledge content:

Your Accessible Flight helps passengers with mobility-related accessibility needs plan travel.

Sydney Airport accessibility services include:

- Wheelchair assistance
- Accessible toilets
- Accessible parking
- Mobility assistance
- Airport staff support

Passengers should contact airlines before travelling to request additional assistance.

After uploading the document, Pinecone processed the knowledge source and the Assistant became ready for chatbot requests.

Final Verification

The chatbot was tested using the current /api/ai/chat endpoint after configuring the knowledge source.

Results obtained from the final testing confirmed the following:

- The requests are properly passed to the existing chatbot flow
- The configured AI provider is Pinecone
- The Assistant can access knowledge sources that have been uploaded
- Provider connectivity problems have been sorted out

Remaining Limitations

- The Pinecone behavior is based on the uploaded knowledge file's quality and coverage.
- More YAF data resources like those concerning airport, airline, equipment, hotels, and reservations may be incorporated in future tasks of Sprint 2.
- This task is limited to the verification of the AI provider connection only.

The chatbot is already set up to use Pinecone Assistant via the existing YAF AI pipeline.